

TITAN molecular prediction report

16 August 2026

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Research-use molecular profile

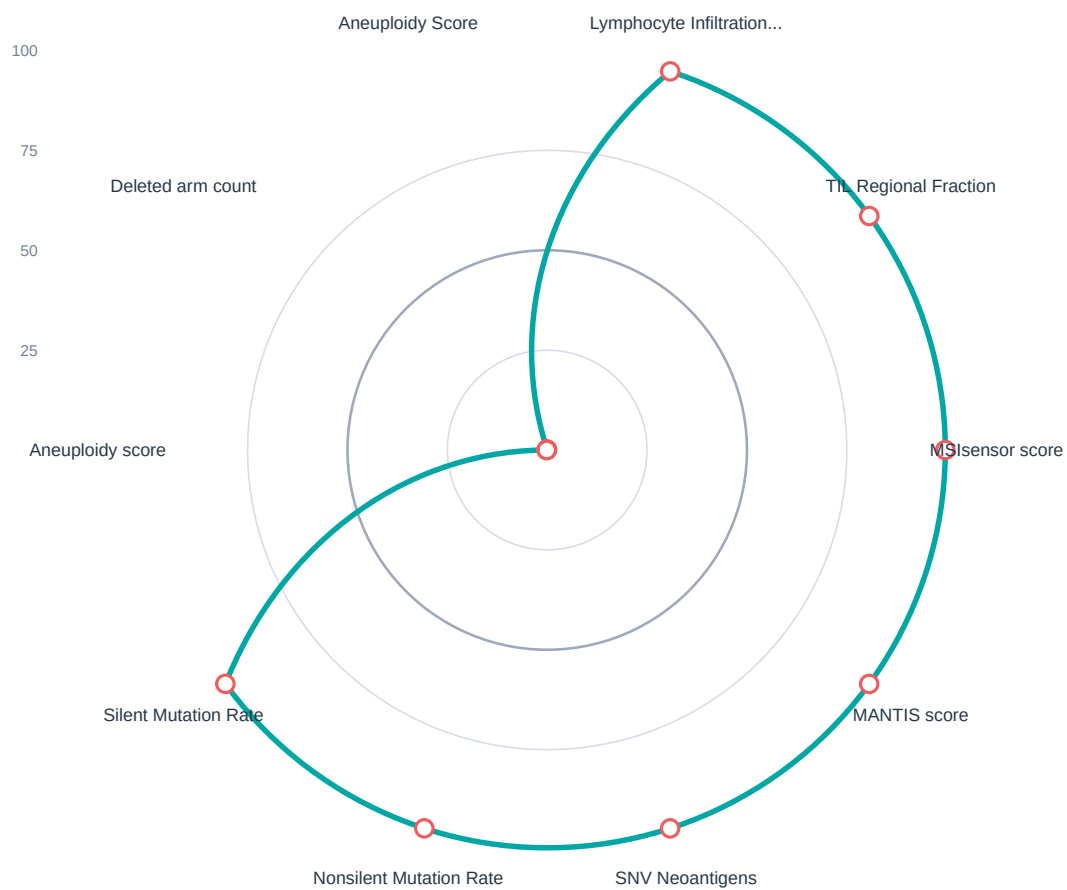
COAD - COAD example A

Models applied	Continuous	Binary	Slides pooled
20	10	10	1

This report applies models trained in TCGA and **not externally validated**. It is a discovery/research output, not a diagnostic report. Binary LDA scores are uncalibrated and must not be read as probabilities.

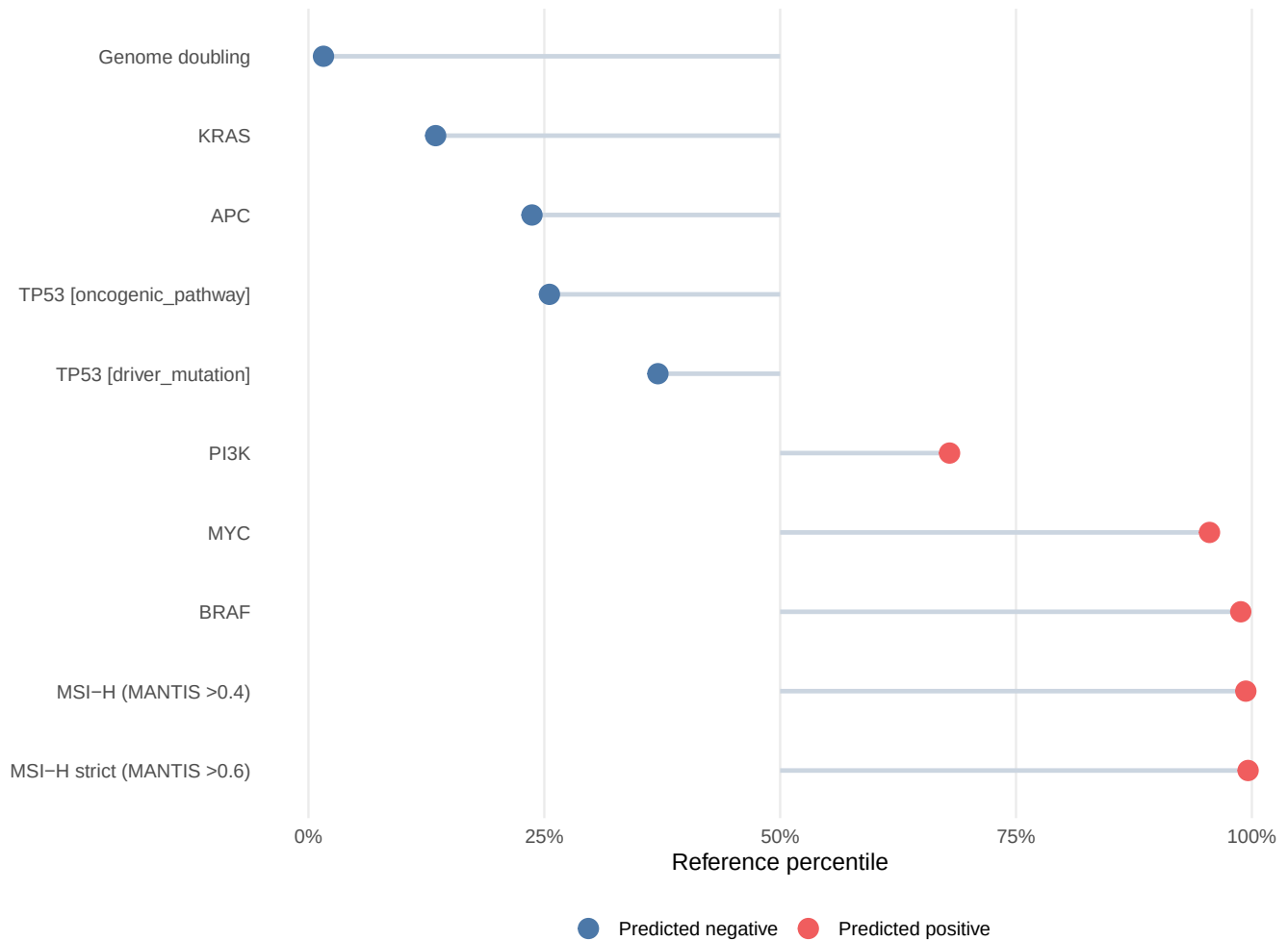
Continuous molecular profile

Percentile relative to repeated TCGA out-of-fold predictions



Binary molecular calls

LDA-score percentile; this is not a calibrated probability



Continuous predictions

Family	Endpoint	Prediction	Reference percentile	Category
aneuploidy	Aneuploidy score	-3.645	0.1%	Moderate effect
aneuploidy	Deleted arm count	-3.754	0.1%	Moderate effect
thorsson	Aneuploidy Score	0.4915	0.1%	Moderate effect
thorsson	Lymphocyte Infiltration Signature Score	0.7478	99.6%	Moderate effect
thorsson	TIL Regional Fraction	14.88	99.6%	Higher effect
microsatellite_instability	MSIsensor score	26.89	99.6%	Moderate effect
microsatellite_instability	MANTIS score	0.9202	99.6%	Moderate effect
thorsson	SNV Neoantigens	7.082	99.6%	Moderate effect
thorsson	Nonsilent Mutation Rate	3.692	99.6%	Moderate effect
thorsson	Silent Mutation Rate	2.652	99.6%	Moderate effect

Binary predictions

Endpoint	Predicted class	LDA score	Score percentile	Category
Genome doubling	0	-5.618	1.6%	Moderate effect
APC	0	-2.257	23.7%	Higher effect
BRAF	1	4.485	98.8%	Moderate effect
KRAS	0	-4.17	13.5%	Moderate effect
TP53 [driver_mutation]	0	-1.476	37.0%	Higher effect
MSI-H (MANTIS >0.4)	1	8.46	99.4%	Higher effect
MSI-H strict (MANTIS >0.6)	1	18.23	99.6%	Higher effect
MYC	1	1.065	95.5%	Moderate effect
PI3K	1	0.4021	68.0%	Moderate effect
TP53 [oncogenic_pathway]	0	-0.1446	25.5%	Moderate effect

Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.0%.
- TITAN input schema: 768 named features (`titan_000`–`titan_767`).
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- Intended use: research only; TCGA discovery atlas without independent external validation.